



Amazon Elastic File System

Lecture Slides

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Introduction to Elastic File System (EFS)



Elastic File System (EFS) is a file storage service for EC2 instances

- Storage capacity grows (up to petabytes) and shrinks automatically based on data stored (elastic)
- **Multiple EC2 instances** in the same VPC **can mount a single EFS Volume** (volume must be in the same VPC)
- EC2 instances install the **NFSv4.1 client** and can then mount the EFS volume
- EFS is using the Network File System version 4 (**NFSv4**) protocol
- EFS creates multiple **mount targets** in all your VPC subnets.
- You based per space used starting at \$0.30 GB / month

VPC	Availability Zone	Subnet	IP address	Mount target ID	Network interface ID	Security groups	Mount target state
vpc-75fcd90f (default)	us-east-1b	subnet-edd727b1 (default)	172.31.36.225	fsmt-1a56eefa	eni-089be3fede4855115	sg-c4b0668c - default	Available
	us-east-1a	subnet-4d1f9b07 (default)	172.31.24.213	fsmt-1c56eefc	eni-07f1538413c5581a7		Creating
	us-east-1c	subnet-9706c1f0 (default)	172.31.10.213	fsmt-1e56eefe	eni-0a685b10dbc4144aa		Creating
	us-east-1e	subnet-cd8e99f2 (default)	172.31.63.220	fsmt-1f56eeff	eni-0120337a18e205006		Creating
	us-east-1d	subnet-ff7684d1 (default)	172.31.85.52	fsmt-e157ef01	eni-043ea2b3a223e4944	sg-c4b0668c - default	Available
	us-east-1f	subnet-fa0e81f5 (default)	172.31.76.141	fsmt-e257ef02	eni-04b4c60064f530831	sg-c4b0668c - default	Available

Amazon EFS Client

amazon-efs-utils package is an **open-source collection of Amazon EFS tools**.
also known as the Amazon EFS client

<https://github.com/aws/efs-utils>

- EFS client enables the ability to use Amazon CloudWatch to monitor an EFS file system's mount status.
- You need to install the Amazon EFS client on an Amazon EC2 instance prior to mounting an EFS file system



includes the Amazon EFS mount helper, which makes it easier to **mount** EFS file systems

mount helper is a program that you use when you mount a specific type of file system

Amazon Linux 2 (AL2) install:

```
sudo yum install -y amazon-efs-utils
```

Amazon EFS Client

EFS mount helper provides the following options:

- Mounting on supported EC2 instances
- Mounting with IAM authorization
- Mounting with Amazon EFS access points
- Mounting with an on-premise Linux client
- Auto-mounting EFS file systems when an EC2 instance reboots
- Mounting a file system when creating a new EC2 instance
- Can mount either Linux or Mac



Amazon EFS **does not support** mounting from Amazon EC2 Windows instances.



Before EFS mount helper standard Linux NFS client was recommended for mounting

Amazon EFS Client

- mount helper defines a new network file system type, called **efs**
 - which is fully compatible with the standard **mount** command in Linux.
- mount helper also supports automating mounting an EFS at instance boot via **/etc/fstab** configuration
- **_netdev** option, used to identify network file systems, when mounting your file system automatically
 - If omitted EC2 instance might stop responding

You can **mount** with the following:

- File system DNS name
- File system ID
- Mount target IP address



```
sudo mount -t efs -o tls fs-example.efs.us-west-2.amazonaws.com:/ /mnt/efs
sudo mount -t efs -o tls fs-12345678:/ /mnt/efs
sudo mount -t efs -o tls,accesspoint=fsap-12345678 203.0.113.25:/ /mnt/efs
```

EFS Management Console will provide the code via the Attach button.

Amazon EFS Client

EFS mount helper will use the **following mount options**:

- **nfsvers=4.1** – used when mounting on EC2 Linux instances
- **nfsvers=4.0** – used when mounting on supported EC2 Mac instances running macOS Big Sur, Monterey, and Ventura
- **rsize=1048576** – Sets the maximum number of bytes of data that the NFS client can receive for each network READ request to 1048576, the largest available, to avoid diminished performance.
- **wsize=1048576** – Sets the maximum number of bytes of data that the NFS client can send for each network WRITE request to 1048576, the largest available, to avoid diminished performance.
- **hard** – Sets the recovery behavior of the NFS client after an NFS request times out, so that NFS requests are retried indefinitely until the server replies, to ensure data integrity.
- **timeo=600** – Sets the timeout value that the NFS client uses to wait for a response before it retries an NFS request to 600 deciseconds (60 seconds) to avoid diminished performance.
- **retrans=2** – Sets to 2 the number of times the NFS client retries a request before it attempts further recovery action.
- **noresvport** – Tells the NFS client to use a new non-privileged Transmission Control Protocol (TCP) source port when a network connection is reestablished. Using the noresvport option helps to ensure that your EFS file system has uninterrupted availability after a reconnection or network recovery event.
- **mountport=2049** – only used when mounting on EC2 Mac instances running macOS Big Sur, Monterey, and Ventura.